BACKTRACK: pilot implementation plan

A learner is stuck on today's lesson and does not know which earlier skill to review. This pilot will test whether a short, targeted route through Khan material helps them return to that lesson.

The March evidence story

A learner initially cannot complete a current mathematics goal. Their route identifies a useful step to investigate. They complete matched Khan practice, then pass fresh supervised destination tasks. A check 7-14 days later tests whether the return holds. A second educator reuses the destination kit. Every link in this chain must have its own evidence source.

Cohort and scope

Planning target: 80-120 eligible secondary learners across two or three cohorts, conditional on partner access, adviser review, permissions, and school schedules. Seek at least two educators so replication is testable. No named school is currently represented as committed.

Begin the evaluated pilot with two destination families: equations with brackets and quadratic equations. The public demo also includes fractions, ratios, and graphs; those samples do not automatically expand the evaluated cohort. Teachers confirm curriculum fit and may reject or revise either. The public examples use algebra content rather than a certified Philippine grade mapping. The product can grow across grades and subjects as reviewed destination kits become available.

Use one primary destination per enrolled learner for the headline analysis. Learners who already pass the baseline are tracked as a separate maintenance group, not counted as recovered. Public visitors and social-media acquisition users remain separate from the formal cohort.

Partner requirements

- An authorized faculty adviser and school decision maker.
- A mathematics teacher able to review routes and parallel assessments.
- A recurring device/connectivity window, including access to Khan and its video player.
- A responsible adult to supervise formal checks and manage learner support.
- A permitted route-to-Khan reporting method and a school-controlled participant key.
- A plan for learners who lack devices, miss sessions, or need support beyond the kit.

Choose the school routine with partners: a remediation period, lab session, advisory block, after-class group, or structured homework plus supervised assessment. A scenario of one or two 20-30-minute windows per week is a planning option, not a universal requirement.

November: readiness and baseline

Week 1 is a readiness gate rather than automatic enrollment. Complete the school agreement, consent process, safeguarding escalation contacts, item review, access tests, and facilitator rehearsal. Verify Khan reporting permissions with a test class and inspect exactly which fields can be exported. Do not upload minors' identifiers into BACKTRACK.

Then orient learners to the route and honesty choices. Run the supervised baseline before exposing return-test solutions. Record who was eligible, invited, permitted, enrolled, and unable to participate. Lock assessment versions and analysis definitions before viewing outcomes.

December: first deployment cycle

Run the first destination cycle using the agreed schedule. Facilitators group learners by needed support rather than publicly labeling ability. Khan assignments provide practice. Record technical failures, missing sessions, and teacher minutes. Freeze substantive route changes within an evaluation cycle; record urgent corrections and which learners saw which version.

Plan around school breaks. Do not treat a holiday as student disengagement. If the follow-up window crosses a closure, prespecify the adjustment and report it.

January: complete the first comparison and revise

Where permitted, use a randomized delayed-start comparison within schools, with both groups receiving regular instruction and equivalent access to Khan practice. Assess both groups before crossover. If this cannot be executed fairly, use a documented feasibility design and restrict causal language.

Review error patterns, support requests, follow-up completion, and school burden. Adjust instructions or routes based on evidence, then version the second kit. Expansion requires a functioning routine, not simply more registrations.

February: second educator and replication

Have a second educator run a reviewed kit using the written guide. Log every question they ask the team, preparation time, and support time. Repeat relevant practice across several weeks. Complete delayed checks early enough to analyze by March.

March: evaluate and hand over

Close data collection in time for quality checks before March 31. Reconcile authorized Khan records, attendance, assessment forms, and cost logs. Produce results with all denominators and missingness. Deliver the reusable kit, facilitator training, maintenance responsibilities, and continuation decision to the institution.

Roles

Team roster supplied by Matthew: Justin Mesias, coach; Matthew Labrador, Paul Recio, and Harry Gomez, students, University of the Philippines Manila. The following responsibilities are proposed and should be assigned by the team: product and technical operations; school coordination and learning design; evaluation and finance. The coach reviews institutional participation, claims, school arrangements, and reporting. A school educator retains assessment and support decisions.

A single session

1. Confirm the learner's current goal and available block.
2. Check a destination or relevant prerequisite.
3. Investigate a deeper step if needed. Do not infer an entire knowledge profile from one answer.
4. Use the short learning stop: one idea, a worked example, and practice. A matching official Khan video can open at its relevant segment, with an option to continue. Teachers assign the original Khan exercises for the repeated school routine.
5. Return for fresh practice evidence. Use a separate supervised assessment for the formal outcome.
6. Pause or take one optional next turn. A learner may stop without a penalty.

Low connectivity and shared devices

Route state is device-local and survives a reload when browser storage works. That is not a full offline product: initial page loading, Khan practice, and video playback require connectivity. The open interactive page can continue without new network requests. Offer short local worked examples, reschedule Khan practice, and distinguish that activity in reporting.

For shared devices, use school-controlled rotation and clear the saved destination after recording permitted evidence. The public tool has no cross-device account sync. Do not rely on it as the institutional system of record. Record the number of learners excluded or delayed by device access instead of quietly removing them from the denominator.

Human support and stop rules

After repeated difficulty, the public flow offers human help rather than endless retries. Facilitators should also intervene for distress, accessibility problems, or gaps outside the authored graph. Record the support without penalizing the learner. Any privacy or safeguarding incident goes to the institution's designated adult under its approved procedure.

Pause cohort expansion if critical mathematics errors remain, consent is incomplete, reporting access is unavailable, or teachers cannot sustain the routine. An unresolved condition is a reason to revise the implementation, not fabricate completion.

Replication output

The final school kit contains a destination map, item bank with answer rationales, Khan resource map, schedule options, assessment forms A/B/C, learner introduction, facilitator guide, reporting dictionary, cost log, and version record. Release scope and content licensing need institutional review before broad distribution.